

IHH Healthcare

LizWorld Update Mobile Number – Functional Specification

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Document Control

Version	Date	Changed By	Comments
0.1	29 October 2025	Jamie Lee	• Initial release
0.2	30 October 2025	Jamie Lee	• Update 1.3 - Work Flow ○ Updated as per 30 October 2025 Update mobile number specs review call to color code the workflow for alternative flow and full flow. • Update 2 - Functional Requirements ○ Updated as per 30 October 2025 Update mobile number specs review call to update FR 2 and FR 4. • Update 4.1.2 Screen Description ○ Updated as per 30 October 2025 Update mobile number specs review call to update Next-Of-Kin First Name and Last Name field description. • Update 4.1.3 Business Logic ○ Updated as per 30 October 2025 Update mobile number specs review call to add validation rule. • Update 4.3.1 Screen flow ○ Updated as per 30 October 2025 Update mobile number specs review call to remove part of screen flow.

Sign off

Please be advised that the current requirements outlined in the Functional Specification take precedence over those defined in the initial scope of work provided at the time the project was awarded. This Functional Specification reflects the most up-to-date and accurate requirements for the project, and all work should be executed in accordance with these revised guidelines to ensure alignment with the project's objectives and deliverables.

Moving forward, project sign-off will be determined based on the criteria outlined in the Functional Specification, instead of the original scope of work.

Approval

The name of accept this deliverable.

Name	Role/Department	Company	Signature	Date
Approver				
Wong Kee Wee	Principal Security Consultant	Azimuth Labs Pte Ltd		30 October 2025
Grimarez Christopher	Business Services Lead	IHH	 Christopher Grimarez Business Services Lead IHH Healthcare Singapore	30 October 2025
Reviewer				
Wong Kee Wee	Principal Security Consultant	Azimuth Labs Pte Ltd		30 October 2025
Vincent Le	Project manager	IHH		30 October 2025
Prepared By				
Jamie Lee	Business Analyst	Azimuth Labs Pte Ltd		30 October 2025

1 Project Overview

1.1 Purpose

The purpose of this functional specification is to define the requirements, workflow, and system behavior for the Update Mobile Number flow in LizWorld. This document outlines how inpatients can securely update their preferred mobile number for receiving WhatsApp OTPs, including the validation process using patient identifiers and Next of Kin details, the handling of successful and failed verification scenarios, and the synchronization of updated contact information across LizWorld and hospital backend systems.

1.2 Overview

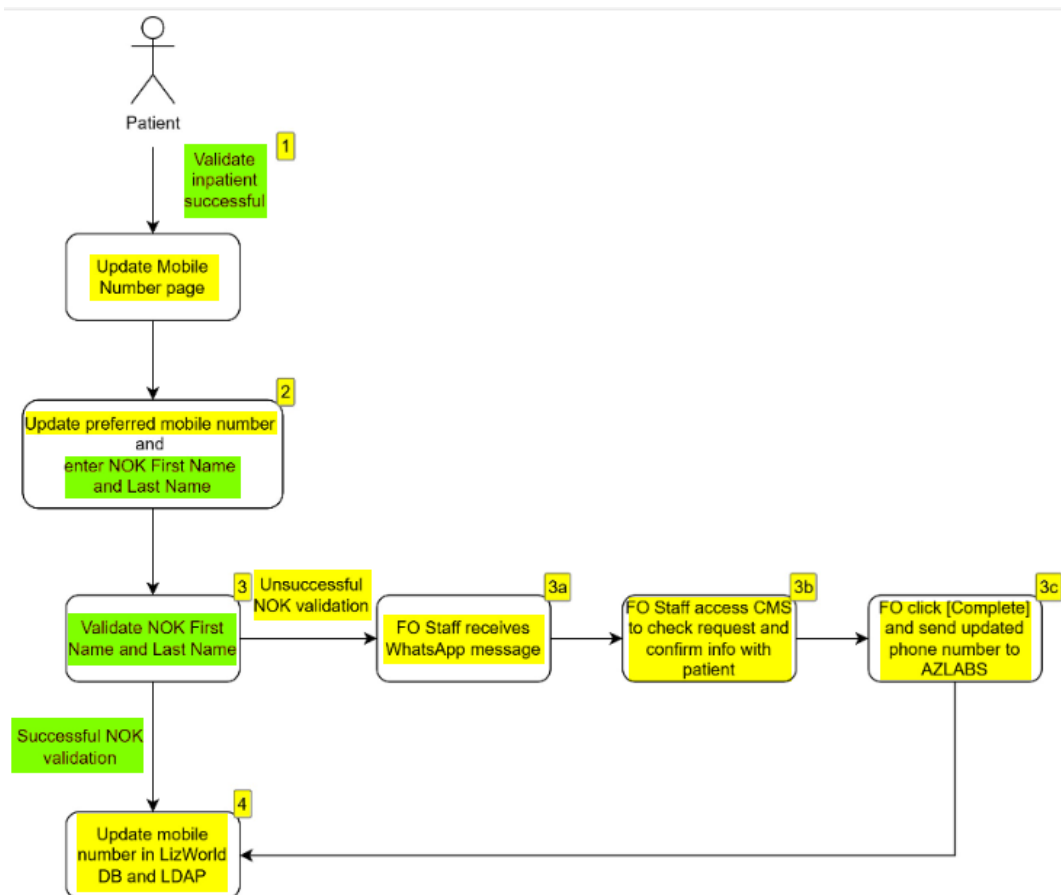
The LizWorld Update Mobile Number flow aims to provide patients with a user self-service option to update their preferred mobile number used for receiving WhatsApp OTPs during login. Currently, the mobile number used for OTP delivery is sourced from ADT messages received from SAP/EAI. However, patients may prefer to use an alternative mobile number for OTP verification during their inpatient stay. The updated mobile number will be stored in a new database table and synchronized across related systems such as SSO, LizWorld DB and CMS, ensuring patients receive OTPs and other service notifications (e.g., limousine booking) on their preferred contact.

Proposed process: The proposed process introduces a user self-service option to update their preferred mobile number. When a patient clicks “Update my mobile” on the OTP page, they will be prompted to scan their hospital wrist tag QR code for validation. With the patient’s ID, Date of Birth, and Case Number from the wrist tag QR code, LizWorld will verify the patient’s inpatient status.

Once validated, the patient can enter their preferred mobile number along with their Next of Kin’s first and last name for verification. If verification succeeds, the new mobile number is stored in a database table, separate from the ADT source data. If verification fails, a WhatsApp notification will be sent to the hospital’s front office staff to manually verify the patient, after which the confirmed mobile number is updated through the CMS system and retrieved by LizWorld.

The updated number is then synchronized across relevant systems and made available for downstream features such as limousine booking, which returns the latest phone number and discharge information to FPT based on the patient’s discharge or estimated discharge date.

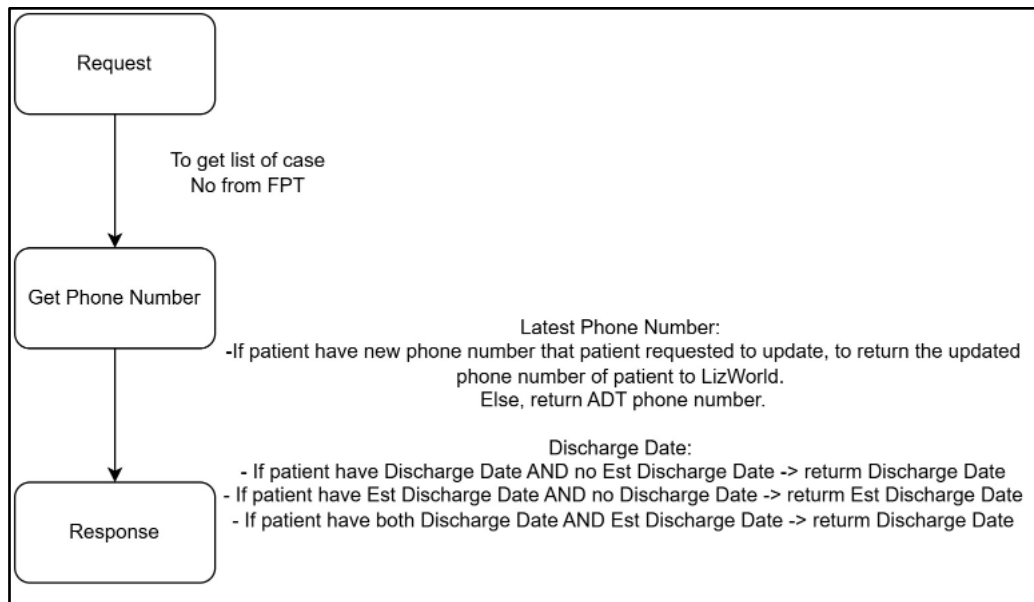
1.3 Work Flow



*Workflow for Update Mobile Number flow
(Yellow = Alternative flow, Green = Full flow)*

Update Mobile Number flow	
(1)	Patient to scan their hospital wrist tag QR code, and LizWorld validates the patient's inpatient status using the Validate API based on Patient ID, Date of Birth, and Case Number. If inpatient validation is successful, the patient proceeds to the Update Mobile Number page.
(2)	On the Update Mobile Number page, the patient enters the preferred mobile number (with country code) and Next of Kin's First and Last Name for verification.
(3)	LizWorld validates the entered NOK details against existing inpatient records.
(3a)	If validation is unsuccessful, the system triggers the manual verification process and a WhatsApp message is automatically sent to the hospital's Front Office (FO) staff to manually verify the patient. (Will be done by FPT)
(3b)	The FO staff access the CMS system to review the request and confirm the patient's information directly with the patient. (Will be done by FPT)

(3c)	Once verified, the FO staff select [Complete] in the CMS to confirm and send the updated mobile number to AZLABS for update processing. (Will be done by FPT)
(4)	For successful NOK validation or after manual confirmation by FO staff, the updated mobile number is: • Stored in the LizWorld database (DB), and • Synchronized with SSO LDAP to maintain data consistency and support future login authentication.



Workflow for LizWorld – Get – Phone Number API

2 Functional Requirements

ID	Requirements	Priority (Refer to Appendix Figure 5.1)
FR 1	Develop the required web pages and pop-ups for the Update Mobile Number flow, including the loading page (after QR scan), error page (validation failure), update mobile number page (with Mobile Number and Next of Kin fields), and verification status pop-ups (completed and in progress).	High
FR 2	Validate inpatient identity via hospital wrist tag QR code scan using the Validate API (Patient ID, DOB, Case No.) and ensure only inpatients in admission where admission date $\leq$ current date $\leq$ discharge date, or admission date $\leq$ current date if discharge date is blank can update their mobile number.	Critical
FR 3	Display an error page if inpatient validation fails, else direct the patient to the Update Mobile Number page to enter the new mobile number and Next of Kin details.	High
FR 4	On successful NOK verification, display the "Verification Completed" pop up page and store the updated mobile number in a database table separate from ADT data. Else, display the "Verification in progress" pop up page.	Critical
FR 5	Send updated contact details (Internal ID, Patient Ward, Patient Name, Patient Mobile, NOK info, request status) to FPT.	High
FR 6	Retrieve verified mobile numbers from CMS and store the updated mobile number in a database table separate from ADT data.	Critical
FR 7	Return the latest phone number and discharge date for patients (based on discharge or estimated discharge date) to FPT.	High

3 Non-Functional Requirements

Detail all non-functional requirements (NFRs) of the project, including such things as performance, usability, security, system behaviour, and project characteristics that relate to user experience.	
ID	Requirement
NFR 1	Only successful validated inpatients can use the update mobile number feature.
NFR 2	Data must be encrypted such tokens to prevent unauthorized access.
NFR 3	All update activities must be logged for traceability and audit purposes.

4 List of User Story

4.1 LizWorld Update Mobile Number web page

User story Overview	This user story allow patient to update mobile number.
Actor	Hospital inpatient
Pre-condition	Patient is not logged in
Trigger	• Patient to access the web page via IPTV QR or direct URL input • Patient to log in from the burger menu • Patient to complete QR code log in or Manual log in • Patient to select "Update my mobile" • Patient to scan their hospital wrist tag QR code.
Post Condition	Patient is presented with the update mobile number web page.

4.1.1 Screen Flow

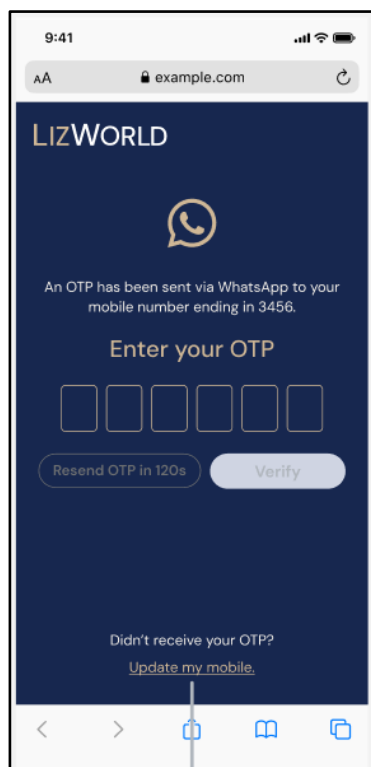


Figure 4.1.1.1: LizWorld WhatsApp OTP Page

The OTP login page will be designed by the FPT team and will be developed and handled by the AZLABS team.



Figure 4.1.1.2: LizWorld QR code scanner

The QR code scanner will be designed, developed and handled by the FPT team.

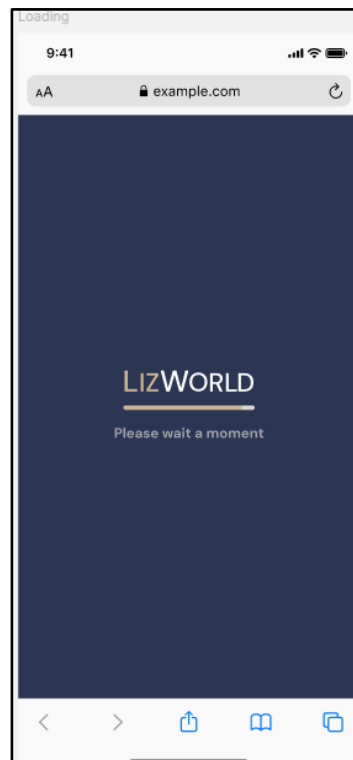


Figure 4.1.1.3: Loading page before Update Mobile Number page

The loading page will be designed by the FPT team and will be developed and handled by the AZLABS team.

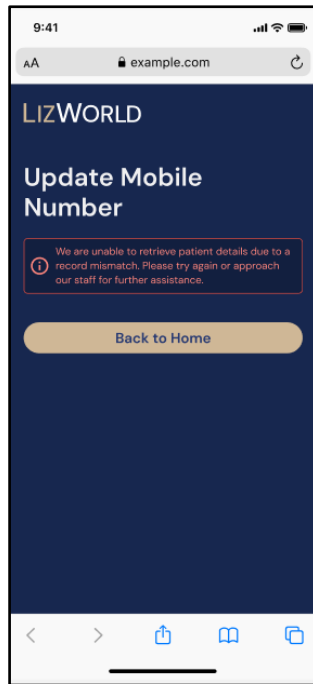


Figure 4.1.1.4: Error page if validation of inpatient fails

The error page will be designed by the FPT team and will be developed and handled by the AZLABS team.

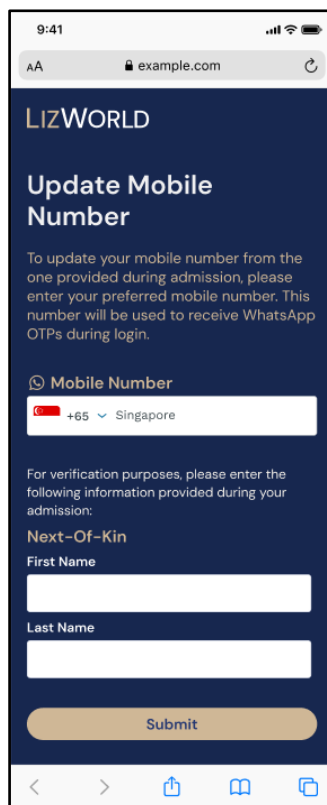


Figure 4.1.1.5: LizWorld Update Mobile Number page if validation of inpatient success

The Update Mobile Number page will be designed by the FPT team and will be developed and handled by the AZLABS team.

4.1.2 Screen Description

Figure 4.1.2.1: LizWorld Update Mobile Number web page

Fields	Field Type	Mandatory/Optional	Max length	Description
Mobile Number	Text field with country code dropdown	Mandatory	N.A.	Allows the patient to enter the preferred mobile number. The country code dropdown is pre-sorted by priority: Singapore, Malaysia, Indonesia, Vietnam, followed by other countries

				in alphabetical order.
Next-Of-Kin First Name	Text field	Optional	N.A.	Field for the patient to enter the Next of Kin's first name for verification purposes. This field is optional and may be left blank if Next of Kin information is not available from ADT.
Next-Of-Kin Last Name	Text field	Optional	N.A.	Field for the patient to enter the Next of Kin's last name for verification purposes. This field is optional and may be left blank if Next of Kin information is not available from ADT.

4.1.3 Business Logic

S/N	Business Rule Name	Details
1.0	Validation Rule	- Validation message will be shown when patient select "Submit" button. - Display red text "Mobile number field cannot be blank." inline message when patient does not input the mobile number mandatory field.
1.0	Action Button	The "Submit" button is designed to submit and trigger the verification process.

4.2 LizWorld Verification status pop up

User story Overview	This user story allow patient to login again if verification is successful, else to return to LizWorld home page if verification is unsuccessful.
Actor	Hospital inpatient
Pre-condition	Patient is not logged in
Trigger	• Patient to complete and submit the Update Mobile Number page
Post Condition	Patient is able to login again with the updated mobile number or return to LizWorld home page.

4.2.1 Screen Flow

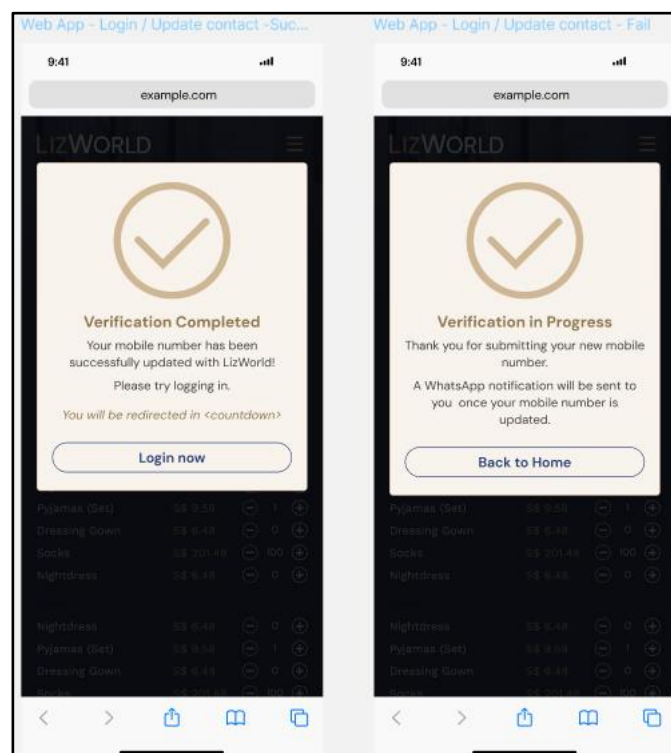


Figure 4.2.1.1: LizWorld Verification Completed and Verification in Progress pop ups

The pop ups will be designed by the FPT team and will be developed and handled by the AZLABS team.

4.2.2 Screen Description

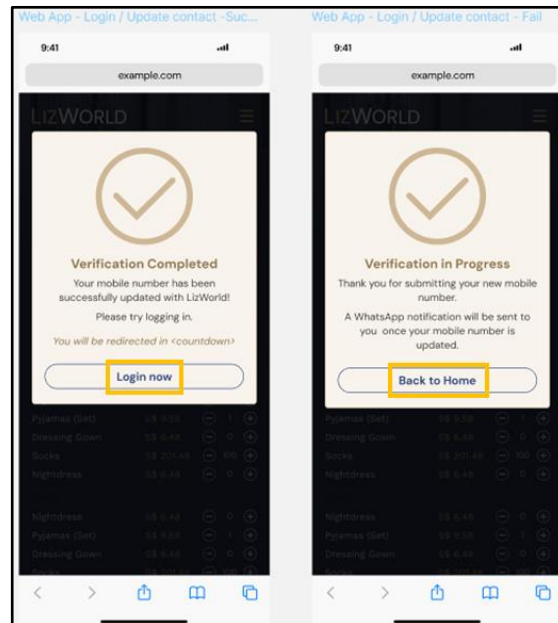


Figure 4.2.2.1: LizWorld Verification Completed and Verification in Progress pop ups

Fields	Field Type	Mandatory/Optional	Max length	Description
Login now button	Navigation control	Mandatory	N.A.	When clicked, it directs the user to the login interface.
Back to home button	Navigation control	Mandatory	N.A.	When clicked, it directs the user to LizWorld home page.

4.2.3 Business Logic

S/N	Business Rule Name	Details
1.0	Action Button	- The "Login now" button is designed to redirect to login page. - The "Back to Home" button is designed to redirect to LizWorld home page.

4.3 LizWorld Update Mobile Number from Front Office

User story Overview	This user story allow front office staff to verify patient's next-of-kin information and check on the updated mobile number in CMS system.
Actor	Front Office staff
Pre-condition	Patient is not logged in
Trigger	• Patient to complete and submit the Update Mobile Number page • Patient's Next-Of-Kin verification is unsuccessful • WhatsApp message is sent to front office staff to assist with the verification
Post Condition	Front office staff will receive a link in the WhatsApp message to proceed with the verification in person and check on the updated mobile number in CMS system.

4.3.1 Screen Flow

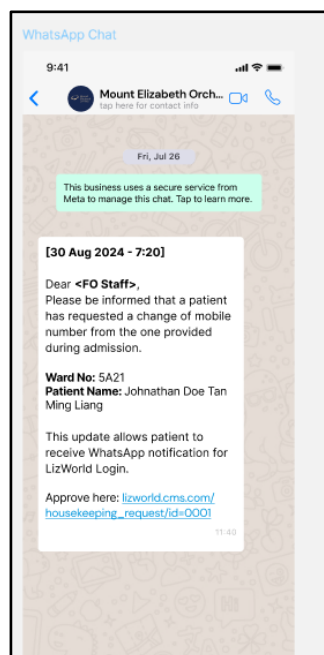


Figure 4.3.1.1: LizWorld Front office staff to verify patient's next-of-kin information and check on the updated mobile number in CMS system.

The WhatsApp notification and web pages will be designed, developed and handled by the FPT team.

4.3.2 Screen Description

This section for LizWorld web page is currently not defined and will be designed, developed, and managed by the FPT team.

4.3.3 Business Logic

This section for LizWorld web page is currently not defined and will be designed, developed, and managed by the FPT team.

5 Appendix

Value	Priority	Description
1	Critical	The requirement is critical to the project's success. Without fulfilling this requirement, the project is not possible.
2	High	The requirement is high priority re the project's success, but the project could still be implemented in a minimum viable product (MVP) scenario.
3	Medium	The requirement is important to the project's success, as it provides value, but the project could still be implemented in an MVP scenario.
4	Low	The requirement is low priority (i.e., it would be nice to have), but the project's success is not dependent upon it.
5	Future	The requirement is outside of the project's scope and is included as a possible component of a prospective release and/or feature.

Figure 5.1: Priority Description with reference to Functional Requirements' priority